

Teaching Exam

Subject – Computer Science

**Topic – Computer Organization &
Architecture**



Lecture No. – 08

By – Rahul Sir

Topics to be Covered

Topic



- 1:- Comparator ✓
- 2:- Types Of Comparator ✓
- 3:- 1,2,4-bit Comparator ✓
- 4:- IC 7485 ✓
- 5:- Application and Conclusion of Comparator ✓
- 6:- computer Hardware ✓
- 7:- Types of Computer Hardware ✓



In digital electronics, a **comparator** is a combinational logic circuit that is used to compare the magnitudes of two binary numbers. Comparators are used in several different electronic circuits like analog to digital converters, voltage level detectors, zero-crossing detectors, etc. The most basic example of a comparator is an XNOR gate which produces a high or logic 1 output only when both its inputs are equal.

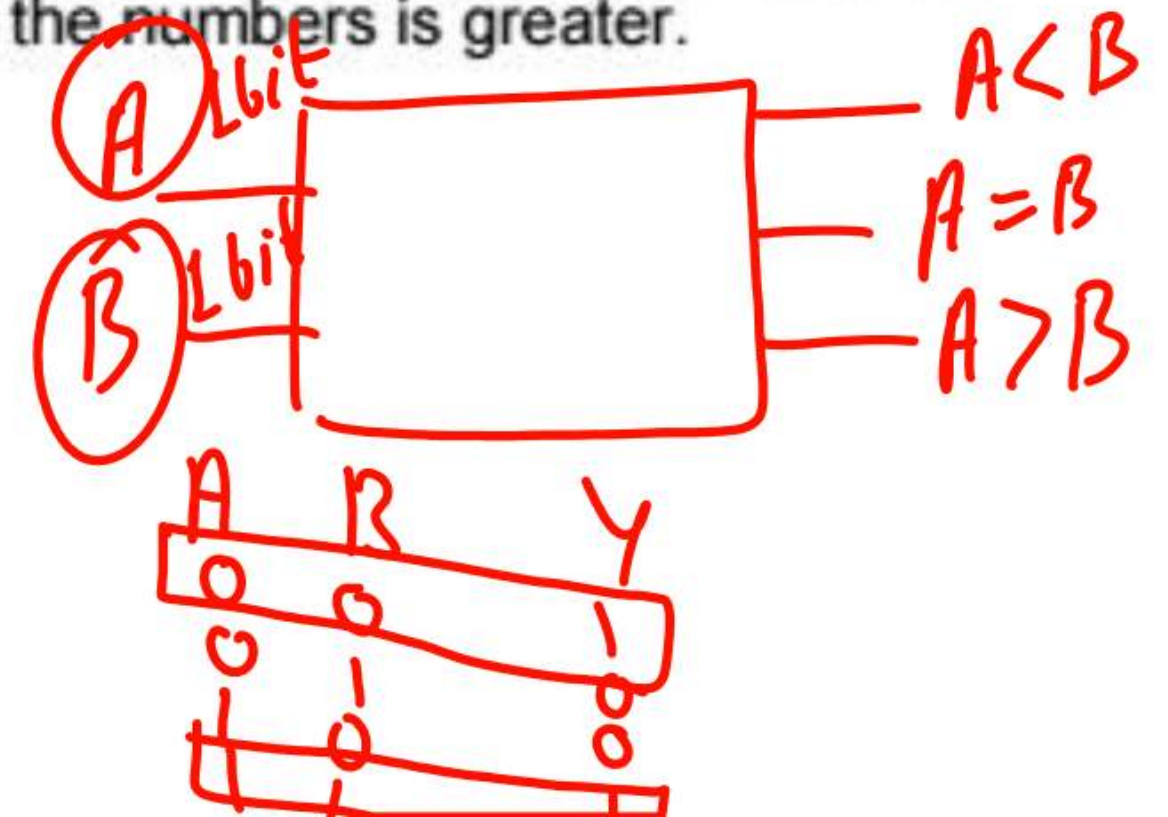
What is a Comparator?

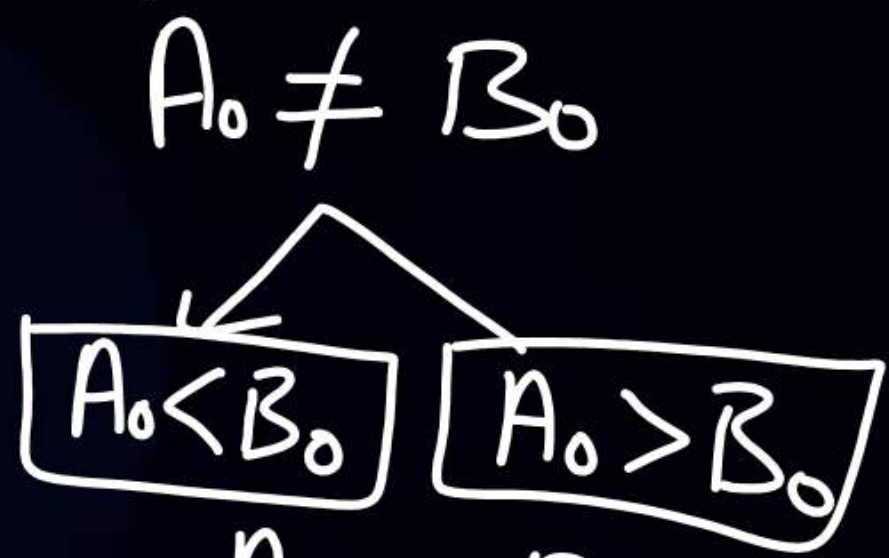
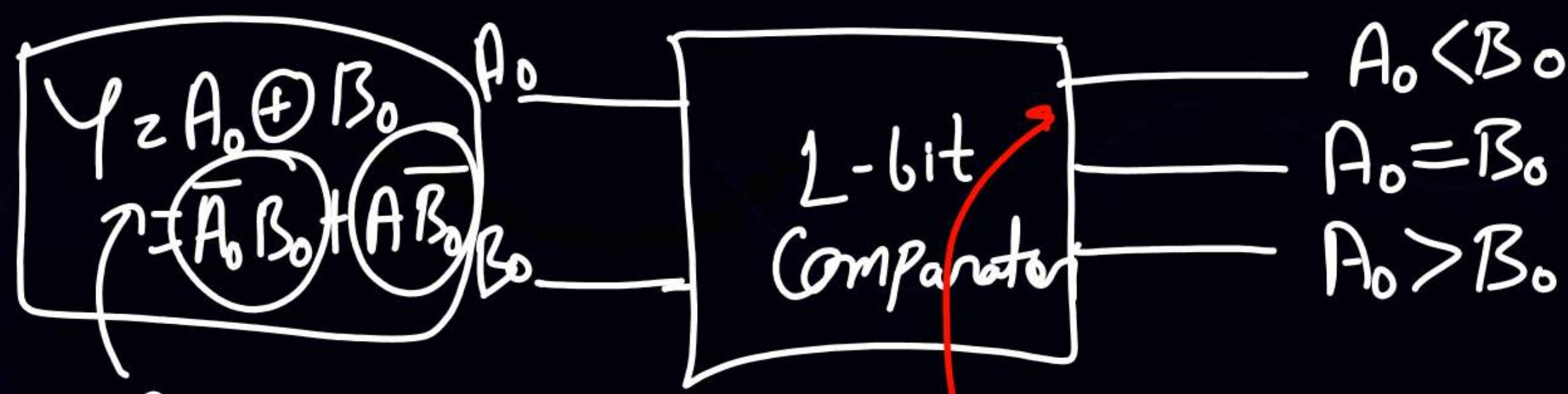
A digital combinational circuit used to compare the magnitude of two binary numbers to determine the equality or non-equality is called a comparator.

Therefore, the main function of a comparator is to compare the values of input numbers and produce an output indicating whether the numbers are equal or specifies which of the numbers is greater.



XNOR





A_0	B_0	Y
0	0	0
0	1	1
1	0	1
1	1	0

* $A_0 < B_0$
 $Y = \bar{A}_0 B_0$

* $A_0 > B_0$
 $Y = A_0 \bar{B}_0$

$Y = \bar{A}_0 \bar{B}_0 + A_0 B_0$
 * $Y = A_0 \odot B_0$
 $A_0 = B_0$



Types of Comparators

Depending on the number of bits, the following are some main types of comparators used in digital circuits –

- 1-Bit Magnitude Comparator
- 2-Bit Magnitude Comparator
- 4-Bit Magnitude Comparator

Let us discuss each type of comparator in detail.

1-Bit Magnitude Comparator

A **1-bit magnitude comparator** is a logic circuit which can compare two binary numbers of one bit each. It produces an output that indicates the relationship between the two input numbers.

In other words, a 1-bit magnitude comparator is one that compares two 1-bit binary numbers and generates an output showing whether one number is equal to or greater than or less than the other.



Topic : 1-bit magnitude Comparator

The block diagram of a 1-bit magnitude comparator is shown in the following figure –



Here, A and B are the 1-bit input numbers, and L, E, and G are the output lines indicating less than or equal to or greater than relationship between A and B respectively.

Let us understand the working this type of comparator.

If $A = 0$ and $B = 0$ or if $A = 1$ and $B = 1$, then $A = B$. It indicates that the two binary numbers are equal. Therefore,

$$E = \bar{A} \cdot \bar{B} + A \cdot B = A \odot B$$

If $A = 0$ and $B = 1$, then $A < B$. This indicates that the binary number A is less than the binary number B. Therefore,

$$L = \bar{A} B$$

If $A = 1$ and $B = 0$, then $A > B$. It indicates that the binary number A is greater than the binary number B. Therefore,

$$G = A \bar{B}$$



Topic : 1-bit magnitude Comparator

Inputs		Outputs		
A	B	L (A < B)	E (A = B)	G (A > B)
0	0	0	1	0
0	1	1	0	0
1	0	0	0	1
1	1	0	1	0

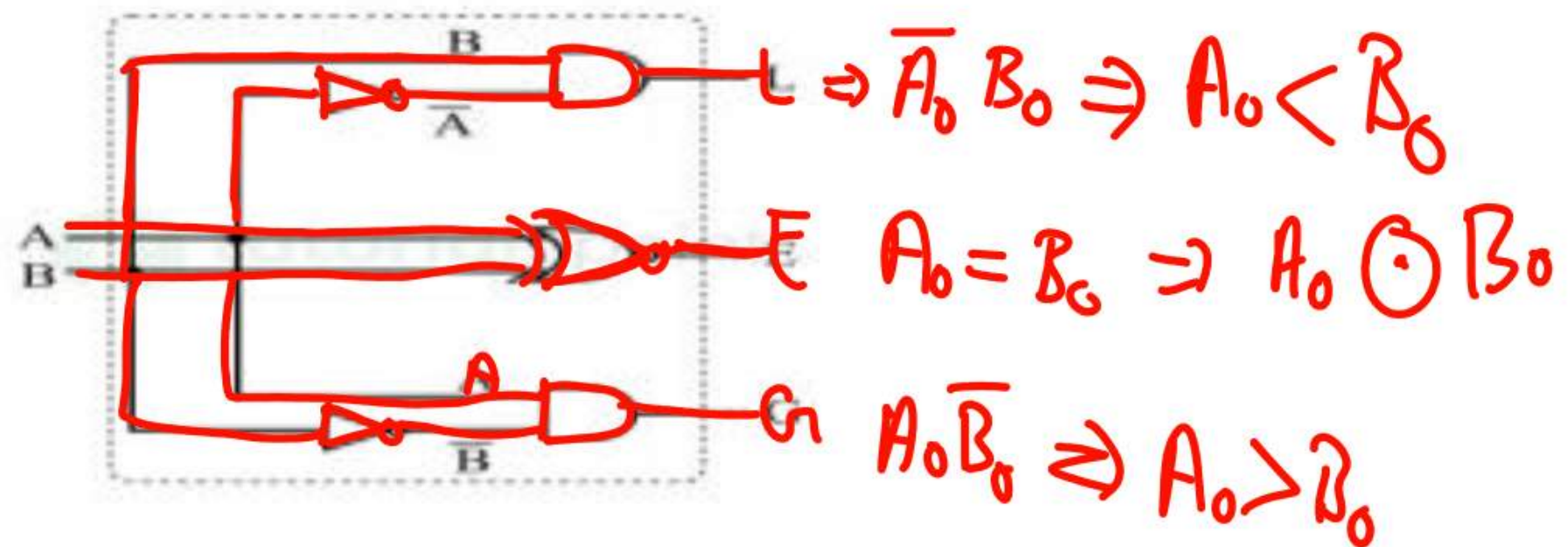
We can use this truth table to obtain the Boolean expression of the 1-bit magnitude comparator.

$$L = \bar{A} B$$

$$E = \bar{A} \cdot \bar{B} + A \cdot B = A \odot B$$

$$G = A \bar{B}$$

The logic circuit diagram of the 1-bit magnitude comparator is shown in the following figure.





Topic : 2-bit magnitude Comparator



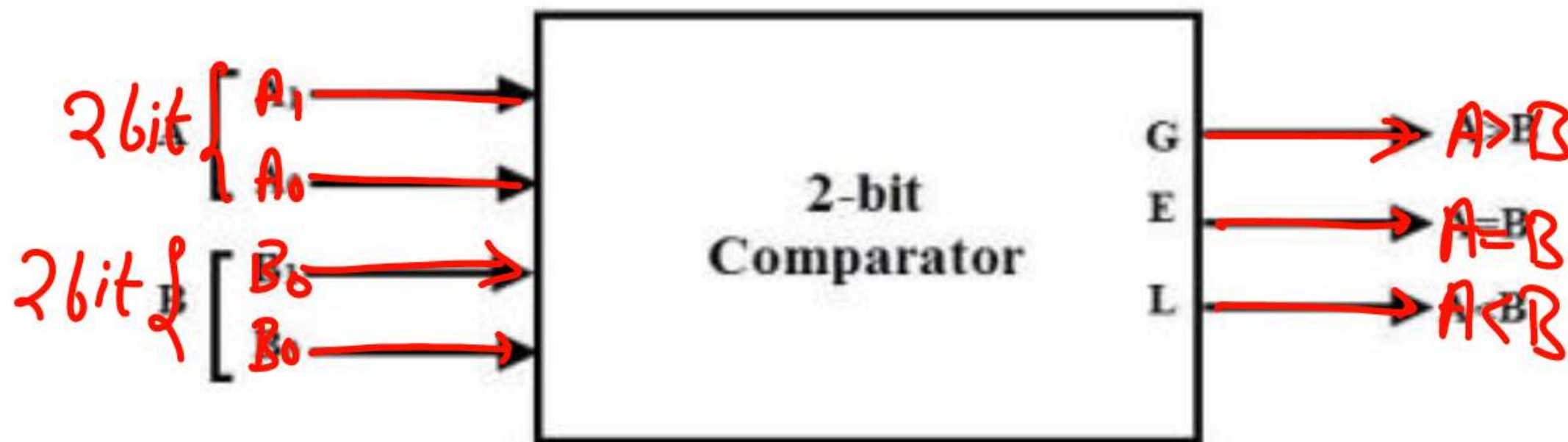
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2-Bit Magnitude Comparator

A digital combinational circuit used to compare the magnitudes of two 2-bit binary numbers and determine the relationship between them is called a **2-bit magnitude comparator**.

Hence, the 2-bit magnitude comparator compares the values represented by two 2-bit binary numbers and then generates an output that indicates whether one number is equal to or greater than or less than the other.

The block diagram of a typical 2-bit magnitude comparator is shown in the following figure –



Here, the lines A_0A_1 and B_0B_1 represents two 2-bit binary number inputs and the lines L, E, and G represents the less than, equal to, and greater than output lines.

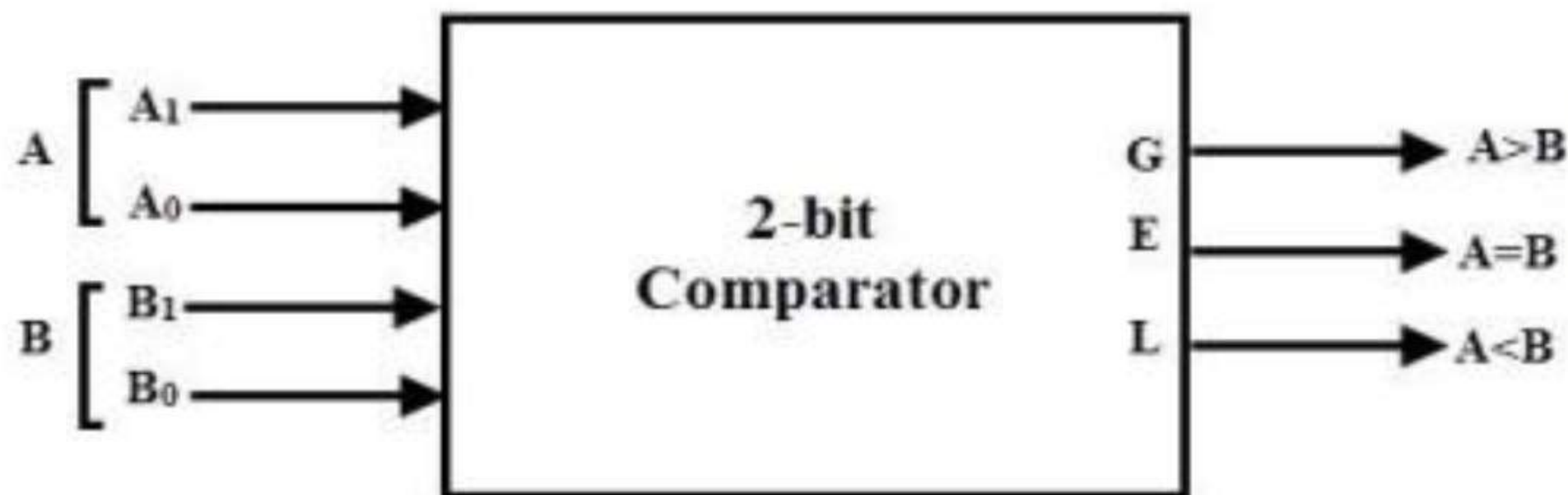


2-Bit Magnitude Comparator

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Topic : 2-bit magnitude Comparator



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Inputs				Outputs		
A_1	A_0	B_1	B_0	$L (A < B)$	$E (A = B)$	$G (A > B)$
0	0	0	0	0	1	0
0	0	0	1	1	0	0
0	0	1	0	1	0	0
0	0	1	1	1	0	0
0	1	0	0	0	0	1
0	1	0	1	0	1	0
0	1	1	0	1	0	0
0	1	1	1	1	0	0
1	0	0	0	0	0	1
1	0	0	1	0	0	1
1	0	1	0	0	1	0
1	0	1	1	1	0	0
1	1	0	0	0	0	1
1	1	0	1	0	0	1
1	1	1	0	0	0	1
1	1	1	1	0	1	0



Topic : 2-bit magnitude Comparator



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Case 1: $A = B$

The comparator produces an output $A = B$ which is E , if $A_0 = B_0$ and $A_1 = B_1$. Therefore, the Boolean expression for the output E will be,

$$E = (A_0 \odot B_0) (A_1 \odot B_1)$$

Case 2: $A < B$

The comparator produces an output $A < B$ which is L , if

- $A_1 = 0$ and $B_1 = 1$, OR
- $A_1 = B_1$ and $A_0 = 0$ and $B_0 = 1$.

$A > B$

$A_1 = 1$ and $B_1 = 0$

OR

$A_1 = B_1$ and $A_0 > B_0$

From these statements, we can write the Boolean expression for the output L as follows –

$A \bar{B}$

$$L = \bar{A}_1 B_1 + (A_1 \odot B_1) \bar{A}_0 B_0$$

Case 3: $A > B$

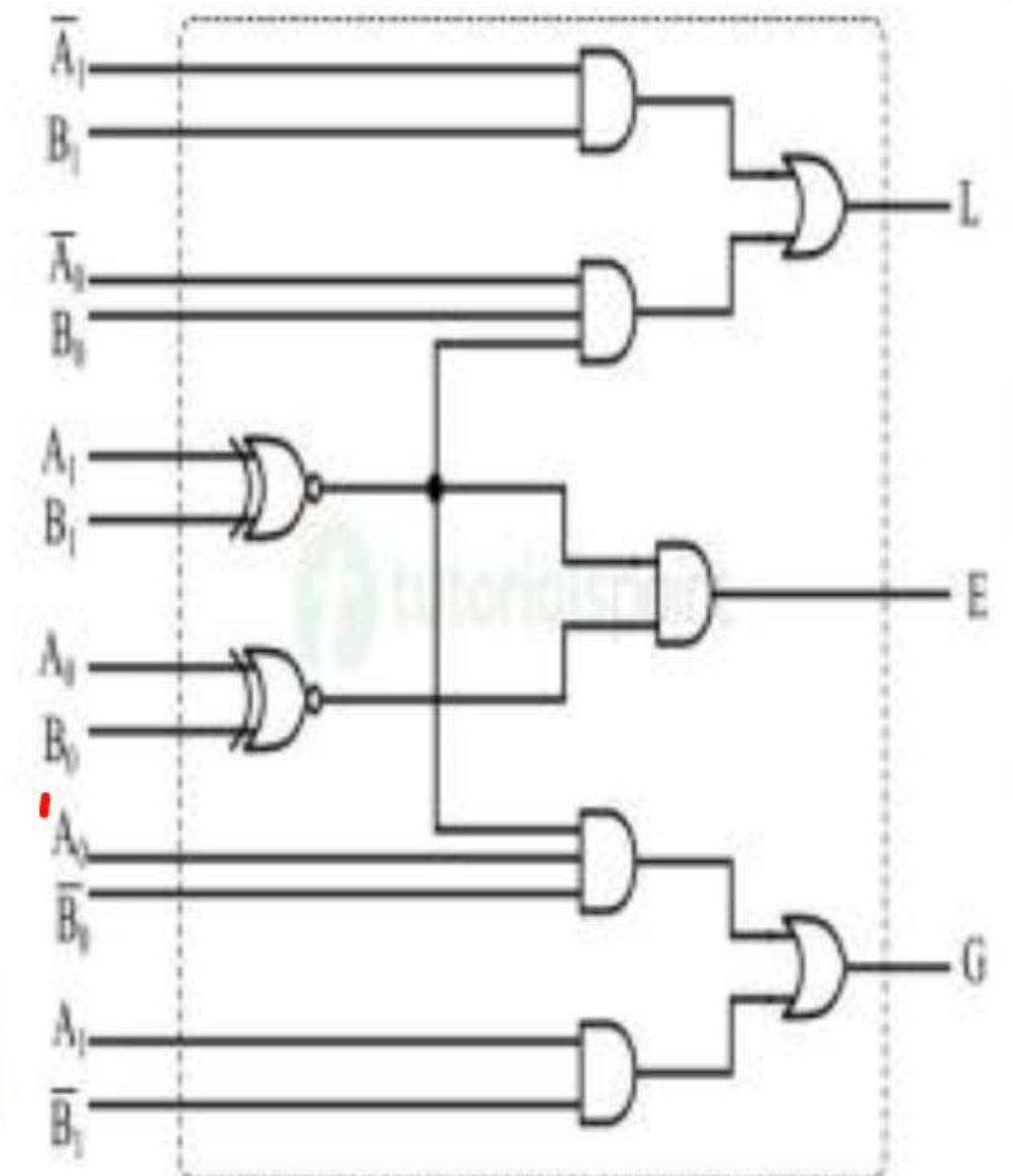
The output of the comparator will be $A > B$ i.e., G , if

- ~~$A_1 = 1$ and $B_1 = 0$~~
- ~~$A_1 = B_1$ and $A_0 = 1$ and $B_0 = 0$~~

$$Y = A_1 \bar{B}_1 + (A_1 \odot B_1) (A_0 \bar{B}_0)$$

From these statements, the Boolean expression for the output G will be,

$$G = A_1 \bar{B}_1 + (A_1 \odot B_1) (A_0 \bar{B}_0)$$





Topic :4-bit magnitude Comparator



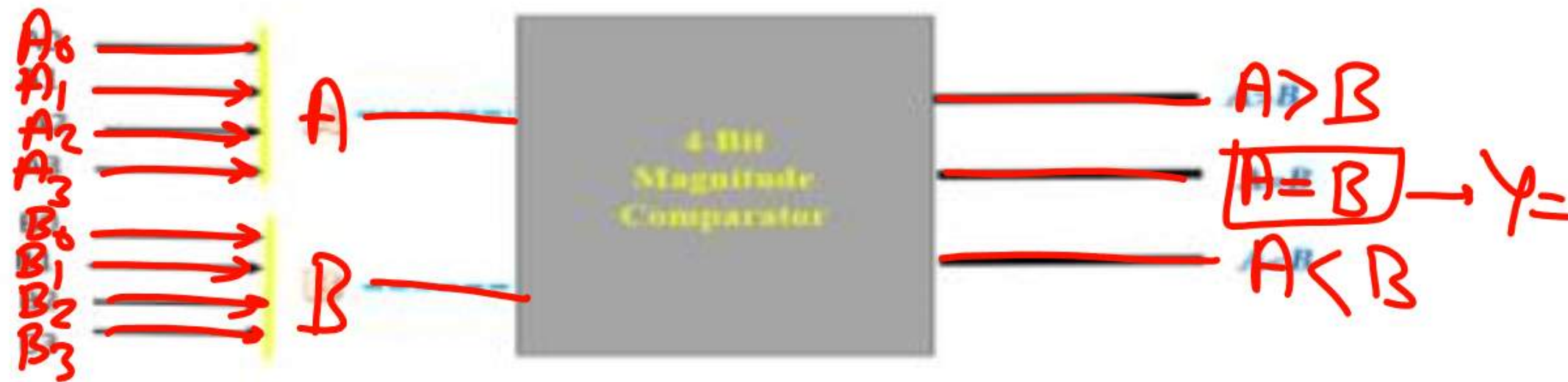
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4-Bit Magnitude Comparator

The 4-bit magnitude comparator is used in more complex digital circuits like microprocessors, microcontrollers, and many more.

It is a type of comparator that can compare the values or magnitudes of two 4-bit binary numbers and produce an output indicating whether one number is equal to or less than or greater than the other.

The block diagram of the 4-bit magnitude comparator is shown in the following figure –



Let us now understand the working of this 4-bit magnitude comparator. For that consider $A = A_3A_2A_1A_0$ is the first 4-bit binary number and $B = B_3B_2B_1B_0$ is the second 4-bit binary number.



Topic : 4-bit magnitude Comparator

Case 1: $A = B$

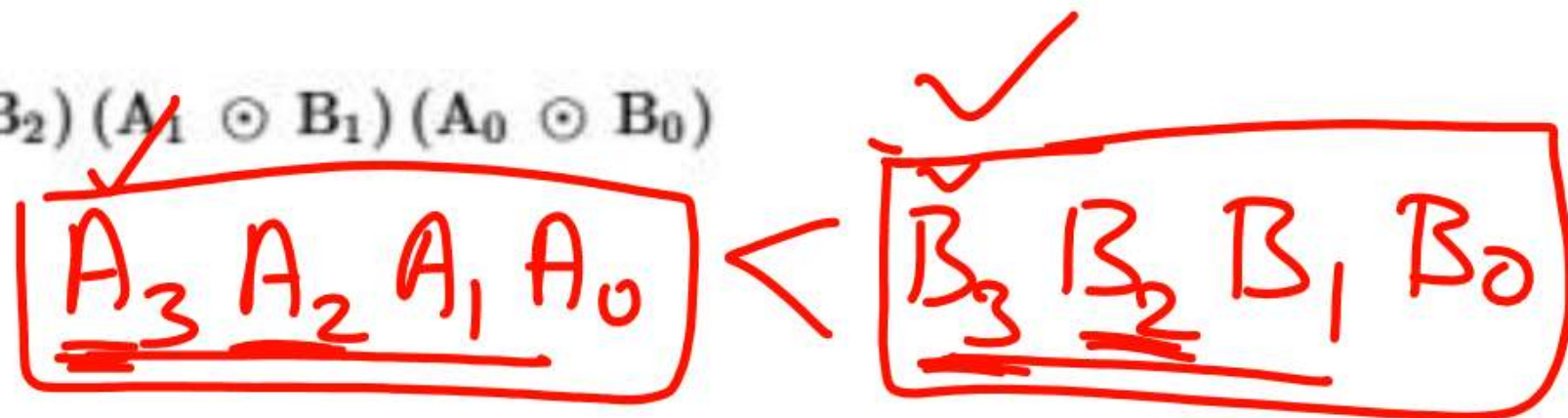
The comparator will produce an output $A = B$ which is E , if all the corresponding bits in the two numbers are equal i.e., $A_3 = B_3$ and $A_2 = B_2$ and $A_1 = B_1$ and $A_0 = B_0$.

In this case, the Boolean expression of the output will be,

$$E = (A_3 \odot B_3) (A_2 \odot B_2) (A_1 \odot B_1) (A_0 \odot B_0)$$

Case 2: $A < B$

$A < B$



The comparator will produce an output $A < B$ which is L , if

- $(A_3 = 0 \text{ and } B_3 = 1)$ OR
- $(A_3 = B_3 \text{ and if } A_2 = 0 \text{ and } B_2 = 1)$ OR
- $(A_3 = B_3 \text{ and if } A_2 = B_2 \text{ and if } A_1 = 0 \text{ and } B_1 = 1)$ OR
- $(A_3 = B_3 \text{ and if } A_2 = B_2 \text{ and if } A_1 = B_1 \text{ and if } A_0 = 0 \text{ and } B_0 = 1)$

$(A_3 = B_3 \text{ and } A_2 = B_2 \text{ and } A_1 = B_1 \text{ and } A_0 = 0 \text{ and } B_0 = 1)$

From these statements, we can derive the Boolean expression for the output L , which is given below.

$$L = \overline{A_3} B_3 + (A_3 \odot B_3) \overline{A_2} B_2 + (A_3 \odot B_3) (A_2 \odot B_2) \overline{A_1} B_1 + (A_3 \odot B_3) (A_2 \odot B_2) (A_1 \odot B_1) \overline{A_0} B_0$$

$$Y = \overline{A_3} B_3 + (A_3 \odot B_3) \overline{A_2} B_2 + (A_3 \odot B_3) (A_2 \odot B_2) \overline{A_1} B_1 + (A_3 \odot B_3) (A_2 \odot B_2) (A_1 \odot B_1) \overline{A_0} B_0$$



Topic :4-bit magnitude Comparator

Case 1: $A = B$

The comparator will produce an output $A = B$ which is E , if all the corresponding bits in the two numbers are equal i.e., $A_3 = B_3$ and $A_2 = B_2$ and $A_1 = B_1$ and $A_0 = B_0$.

In this case, the Boolean expression of the output will be,

$$E = (A_3 \odot B_3) (A_2 \odot B_2) (A_1 \odot B_1) (A_0 \odot B_0)$$

Case 2: $A < B$

The comparator will produce an output $A < B$ which is L , if

- $A_3 = 0$ and $B_3 = 1$, OR
- $A_3 = B_3$ and if $A_2 = 0$ and $B_2 = 1$, OR
- $A_3 = B_3$ and if $A_2 = B_2$ and if $A_1 = 0$ and $B_1 = 1$, OR
- $A_3 = B_3$ and if $A_2 = B_2$ and if $A_1 = B_1$ and if $A_0 = 0$ and $B_0 = 1$.

From these statements, we can derive the Boolean expression for the output L , which is given below.

$$L = \overline{A_3} B_3 + (A_3 \odot B_3) \overline{A_2} B_2 + (A_3 \odot B_3) (A_2 \odot B_2) \overline{A_1} B_1 + (A_3 \odot B_3) (A_2 \odot B_2) (A_1 \odot B_1) \overline{A_0} B_0$$



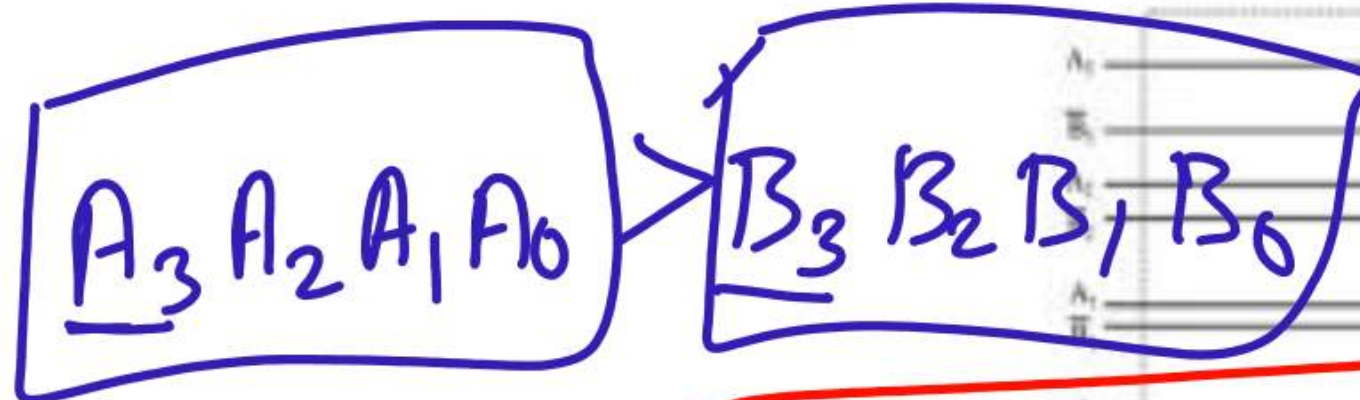
Topic : 4-bit magnitude Comparator



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Case 3: $A > B$

The comparator produces an output $A > B$ which is G , if



$(A_3=1 \text{ and } B_3=0)$ OR

$[(A_3=B_3) \text{ and } A_2 > B_2]$ OR

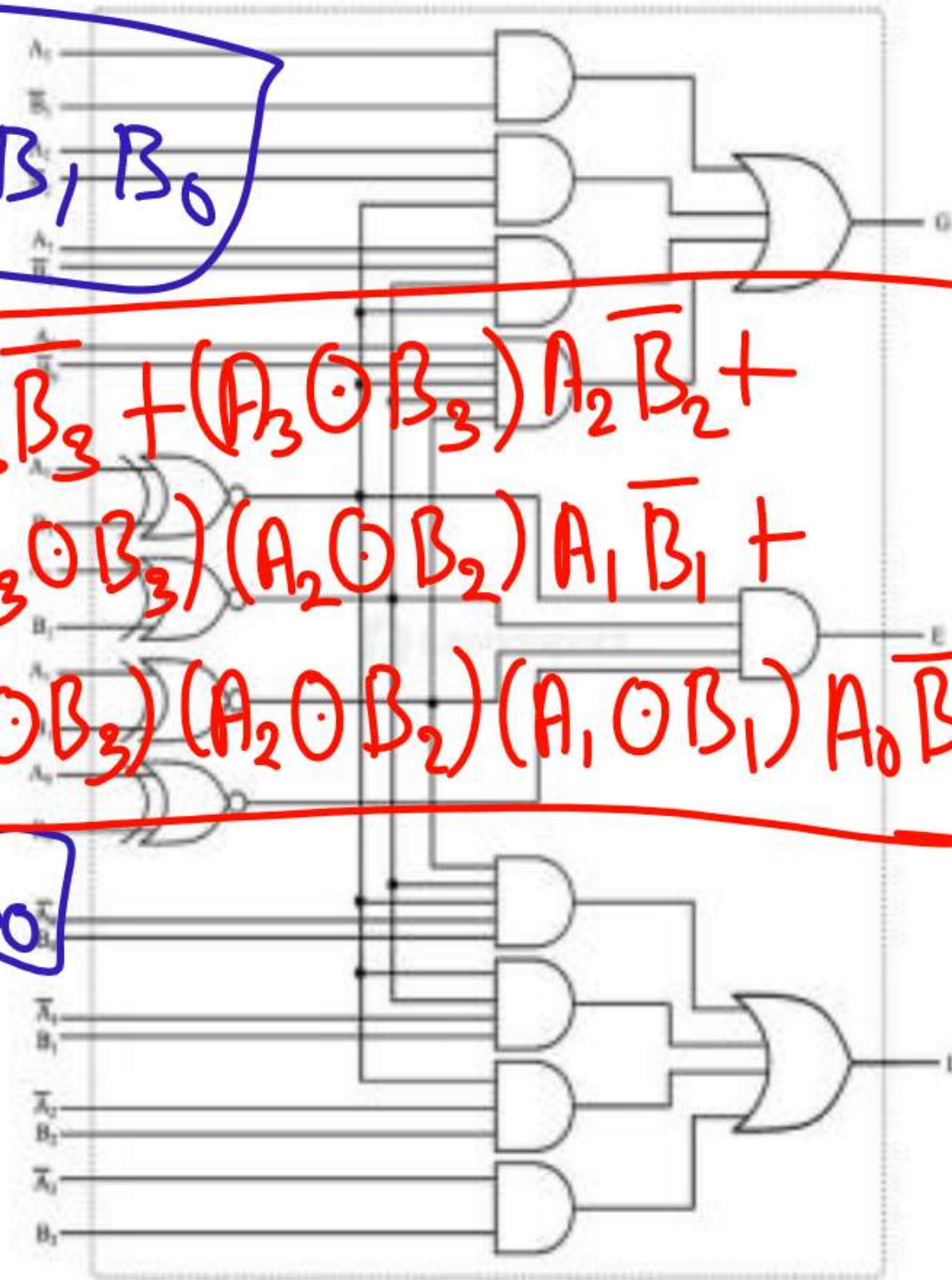
$[(A_3=B_3) \text{ and } (A_2=B_2) \text{ and } A_1 > B_1]$ OR

$[(A_3=B_3) \text{ and } (A_2=B_2) \text{ and } (A_1=B_1) \text{ and } A_0 > B_0]$

Hence, from these statements, we can write the Boolean expression for the output G which is,

$$G = A_3 \bar{B}_3 + (A_3 \odot B_3) A_2 \bar{B}_2 + (A_3 \odot B_3) (A_2 \odot B_2) A_1 \bar{B}_1 + (A_3 \odot B_3) (A_2 \odot B_2) (A_1 \odot B_1) A_0 \bar{B}_0$$

$$Y = A_3 \bar{B}_3 + (A_3 \odot B_3) A_2 \bar{B}_2 + (A_3 \odot B_3) (A_2 \odot B_2) A_1 \bar{B}_1 + (A_3 \odot B_3) (A_2 \odot B_2) (A_1 \odot B_1) A_0 \bar{B}_0$$





Topic :4-bit magnitude Comparator



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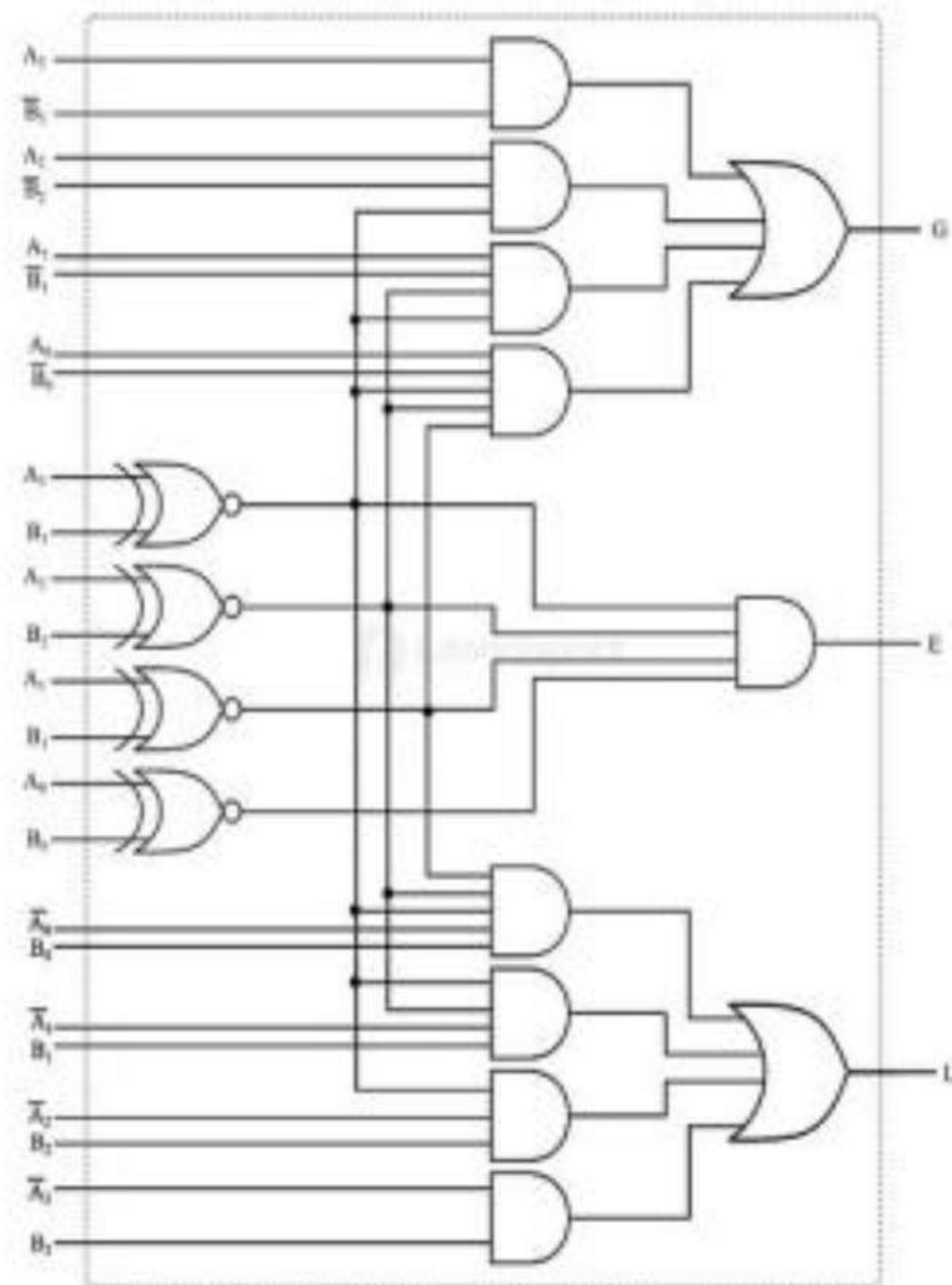
Case 3: $A > B$

The comparator produces an output $A > B$ which is G , if

- $A_3 = 1$ and $B_3 = 0$, OR
- $A_3 = B_3$ and if $A_2 = 1$ and $B_2 = 0$, OR
- $A_3 = B_3$ and if $A_2 = B_2$ and if $A_1 = 1$ and $B_1 = 0$, OR
- $A_3 = B_3$ and if $A_2 = B_2$ and if $A_1 = B_1$ and if $A_0 = 1$ and $B_0 = 0$.

Hence, from these statements, we can write the Boolean expression for the output G which is,

$$G = A_3 \bar{B}_3 + (A_3 \odot B_3) A_2 \bar{B}_2 + (A_3 \odot B_3) (A_2 \odot B_2) A_1 \bar{B}_1 + (A_3 \odot B_3) (A_2 \odot B_2) (A_1 \odot B_1) A_0 \bar{B}_0$$





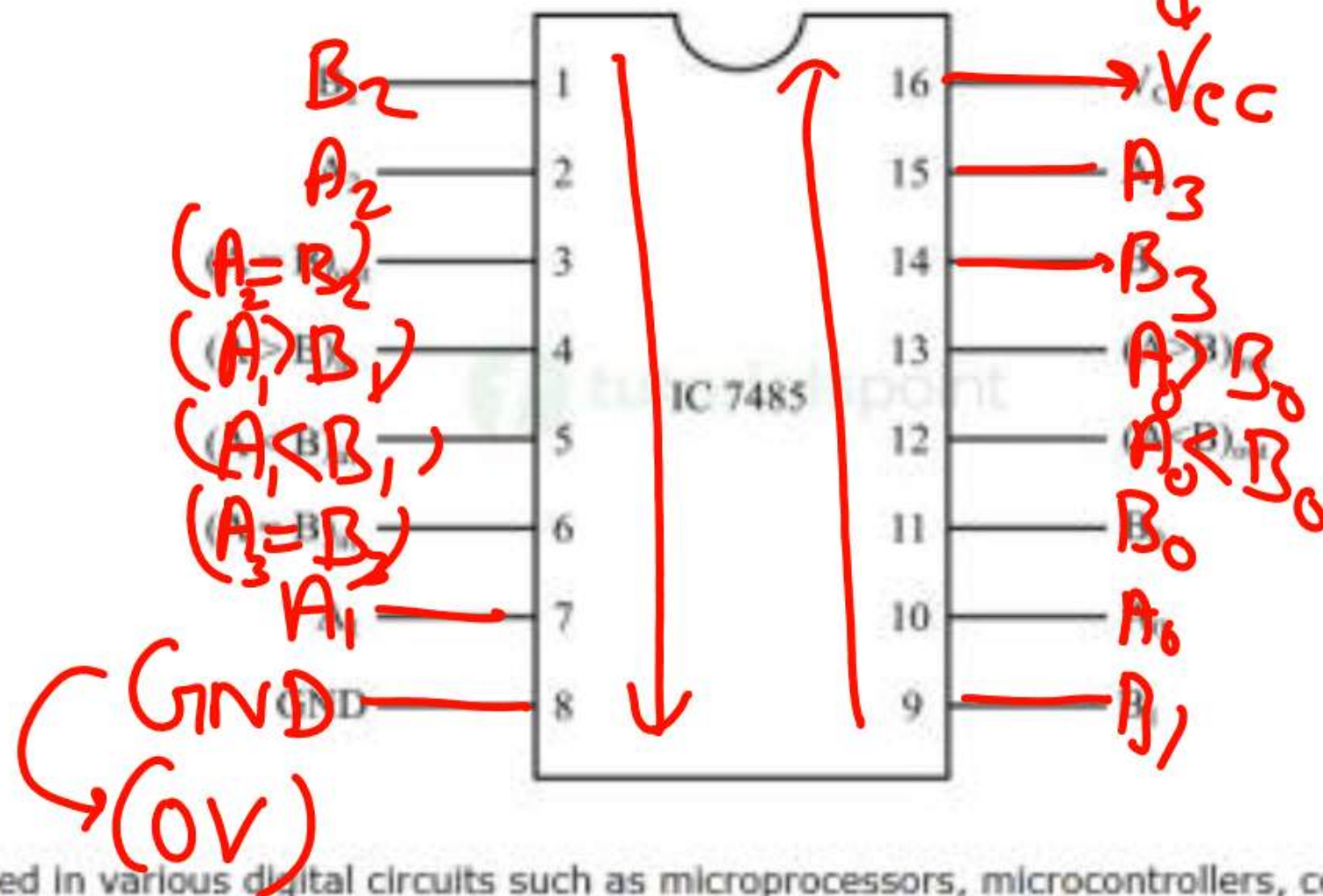
Topic : 4-bit magnitude Comparator IC

Comparator IC

A comparator IC is an integrated circuit which is designed to compare two binary numbers and produce an output based on the comparison results.

A commonly used comparator IC is IC 7485 which is a 4-bit magnitude comparator IC. It is widely in digital electronic applications to compare two 4-bit binary numbers.

The IC 7485 compares corresponding bits of the two input numbers and determines whether one number is equal to or greater than or less than the other. The pin diagram of the IC 7485 is depicted in the following figure.



IC 7485

↳ 4 bit Comparator

This IC is widely used in various digital circuits such as microprocessors, microcontrollers, control systems, and arithmetic logic units.



Applications of Comparators

The comparators are fundamental components in various digital circuits. They provide capabilities to compare voltage levels and make decisions.

Some of the key applications of comparators in the field of digital electronics are listed below –

- Comparators are used to detect changes in voltage levels in electronic circuits.
- Comparators are also used in temperature monitoring systems.
- Comparators are used as zero-crossing detector in various power circuits like phase control circuits, motor and power control circuits, etc.
- Comparators are key components in analog to digital converters (ADCs). ADC
- In signal conditioning circuits, the comparators are used for amplification and filtering of signals before processing.

Conclusion

In conclusion, a comparator is a combinational logic circuit used in various digital electronic applications to compare two similar quantities like two 2-bit binary numbers or two voltage levels, etc.

Comparators help us to make decisions based on a comparison of two input values. They are widely used in several different electronic devices and systems such as arithmetic logic units, microprocessor-based systems, control systems, automation systems, and telecommunication systems.

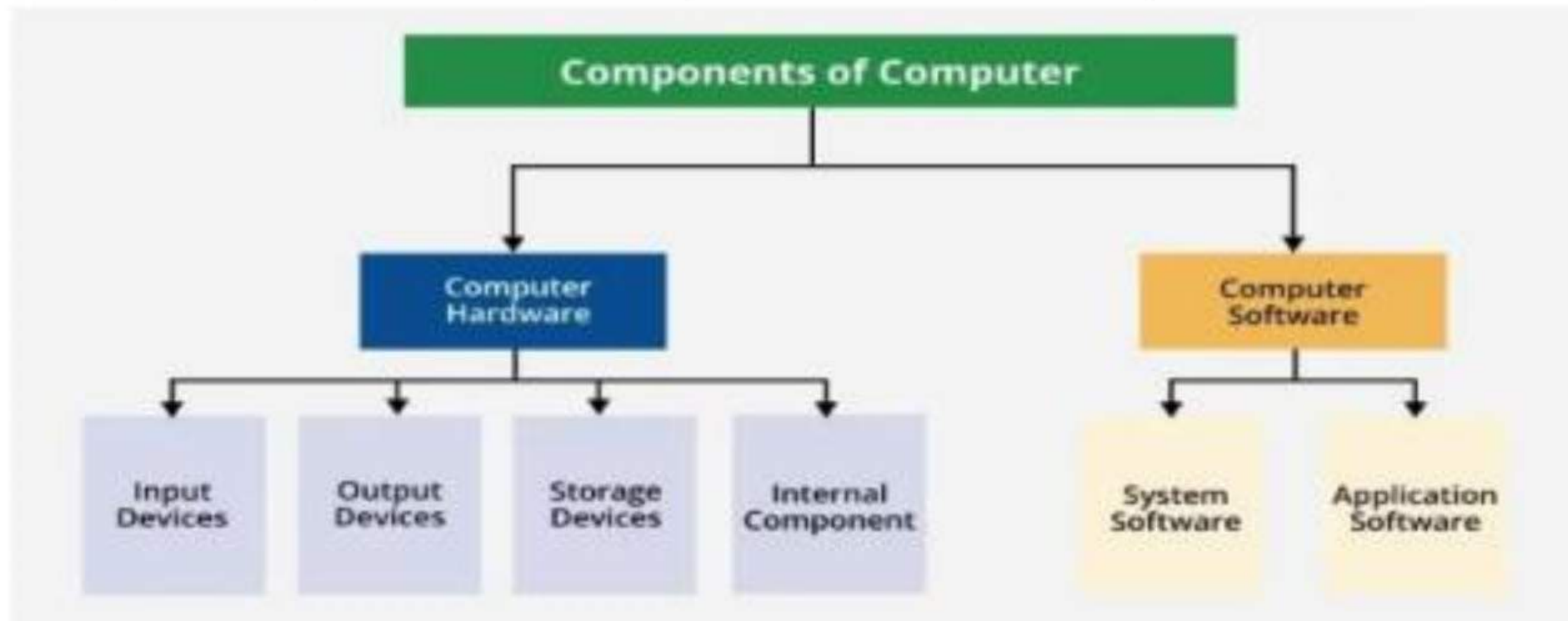
Depending on the number of input bits required for a particular application, we can design a 1-bit, 2-bit, or 4-bit comparator. Also, comparators are available in the form of integrated circuit. The most commonly used IC comparator is IC 7485 which is a 4-bit magnitude comparator.



Computer hardware refers to the **physical components** of a computer that you can see and touch. These components work together to process input and deliver output based on user instructions. In this article, we'll explore the different types of computer hardware, their functions, and how they interact to make your computer work.

What is a Computer Hardware

Computer hardware is a physical device of computers that we can see and touch. E.g. Monitor, Central Processing Unit, Mouse, Joystick, etc. Using these devices, we can control computer operations like input and output.





Keyboard: The most widely used input device, featuring 104 keys, including alphabetic, numeric, and function keys. Modern keyboards connect via Bluetooth, replacing traditional wired connections.

Mouse: A pointing device that controls the cursor on the screen. It features left, right, and middle buttons for selection and interaction. The sensor inside the mouse detects its movement speed, adjusting the cursor accordingly.

Scanner: Scans documents, images, and other media, converting them into digital formats for editing or processing, similar to a Xerox machine.

Microphone: Converts sound into electrical signals. It captures voice input for speech recognition and voice commands on the computer.

Optical Character Reader (OCR): Scans printed or handwritten text, converting it into digital data by detecting reflected light from the characters, similar to a scanner.

Bar Code Reader: Reads bar codes and converts them into digital data for processing. The bar code consists of light and dark lines that encode information.



- **Monitor:** The main output device. It is also called VDU (visual display unit) and it looks like a TV screen. The Monitor displays the information from the computer. It is used to display text, video, images, etc.
- **Printer:** A printer is an output device that transfers data from the computer in a printed format by using text or images on paper. There are both coloured and black & white printers. Further, there are also different types of printers, like Laser Printer, Dot-matrix printers, and Inkjet printers.
- **Plotter:** It is similar to a printer but plotters are large. A plotter is used to generate large drawings, architectural blueprints, etc. on paper and these are high-quality images and drawings and large.
- **Speakers:** It is a very common output device and it gives sound as an output. The speaker is generally used to play music or anything having sound.



Topic: Storage Devices

Some devices are used for storage purposes and are known as secondary storage devices. Some of them are discussed below:

1. CD (Compact disc): A CD is circular and made up of thin platted glass and plastic polycarbonate material. It has a storage capacity of 600 MB to 700 MB of data. It has a standard size of 12 cm with a hole in the centre of about 1.5 cm and 1.2 mm in thickness. There are 3 types of CDs, which are:

- **CD-ROM (CD - Read Only Memory):** Contents of this type of CD cannot be erased by the user. Only the publisher is allowed to access the data imprinted on this CD. CD-ROM is used for commercial purposes like for a music album or any application package by a software company.
- **CD-R (CD-Recordable):** In this, content or data can be stored once. After that, they can be read many times but the data or content cannot be rewritten or erased. (Kind of one-time use)
- **CD-RW (CD-Rewritable):** As the name suggests, this type of CD is used to rewrite the content or erase previous content and again write new content many times.



2. DVD (Digital Video/Versatile Disc): A DVD is the same as a CD but with some more features. A DVD comes in single and dual-layer formats. It has much greater storage capacity in comparison to CD. The storage capacity of a DVD with a one-sided single layer is - 4.7 GB, one-sided double layer - 8.5 GB, double-sided single layer - 9.4 GB, and double-sided double layer - 17 GB. There are also some types of DVDs, which are :

- **DVD-ROM:** In this type, the contents of the DVD cannot be written on or erased by the user. DVD ROM is used for applications and databases for distributing them in large amounts.
- **DVD-RAM:** DVD RAM is accessed like a hard disk. It provides high data security and storage capacity. This is a kind of rewritable disc and it allows up to 1,00,000 rewrites.



3. Hard Disk An hard disk is a non-volatile storage device that uses its read/write heads to store digital data on the magnetic surface of a rigid plate. It is generally 3.5 inches in size for desktops and 2.5 inches in size for laptops. A hard disk can be classified further into 3 types, which are:

- **Internal Hard Disk:** It has a common storage capacity stated as GB or TB. A system case or cabinet is the place where it is located. It can perform faster operations and its storage is fixed. It is mainly used to store large data files and programs.
- **Hard Disk Packs** are used by organizations such as banks and government sector organizations to store large amounts of data. They have a storage capacity of PB (peta bytes).

THANK YOU